Wildfire Project Milestone 3

Exploratory Data Analysis & Preprocessing

ABSTRACT

Transitioning from the 244-row Algerian dataset to a comprehensive 14,988-row California weather and fire-start record (1984–2025). This milestone documents the analysis window, seasonal insights, and a leak-free chronological preprocessing pipeline.

A larger, relevant dataset

How does the new source compare to the initial project scope?

The project has transitioned to the **California Weather and Fire Prediction Dataset (1984–2025)**, solving the "small dataset" constraint identified in the initial session.

This dataset integrates daily meteorological observations from NOAA with wildfire occurrence records from CAL FIRE, providing a robust time-series for analysis.

The analysis window is constrained to **1984–2023** (14,610 days) to ensure consistent label coverage across the record.

Metric	Algerian (Old)	California (New)
Observations	244 rows	**14,988 rows (~60x)**
Time Span	1 season (2012)	**41 Years**
Frequency	Daily	Daily
Target Positives	137	**4,971 (34.0%)**
Numeric Features	12	11 (base) + Categorical

Pedagogical Note: Sample size enables rigorous chronological splitting.

Data quality and coverage decision

Is the raw data ready for modeling "as-is"?

Quality Metrics

Total Raw Rows	14,988
Duplicate Rows	0
Missing Avg Wind	3 records
Missing Target	0

Analysis Window (1984–2023)

Retained Rows	14,610
Excluded Rows	378
Positive Labels	4,971
Positive Rate	34.02%

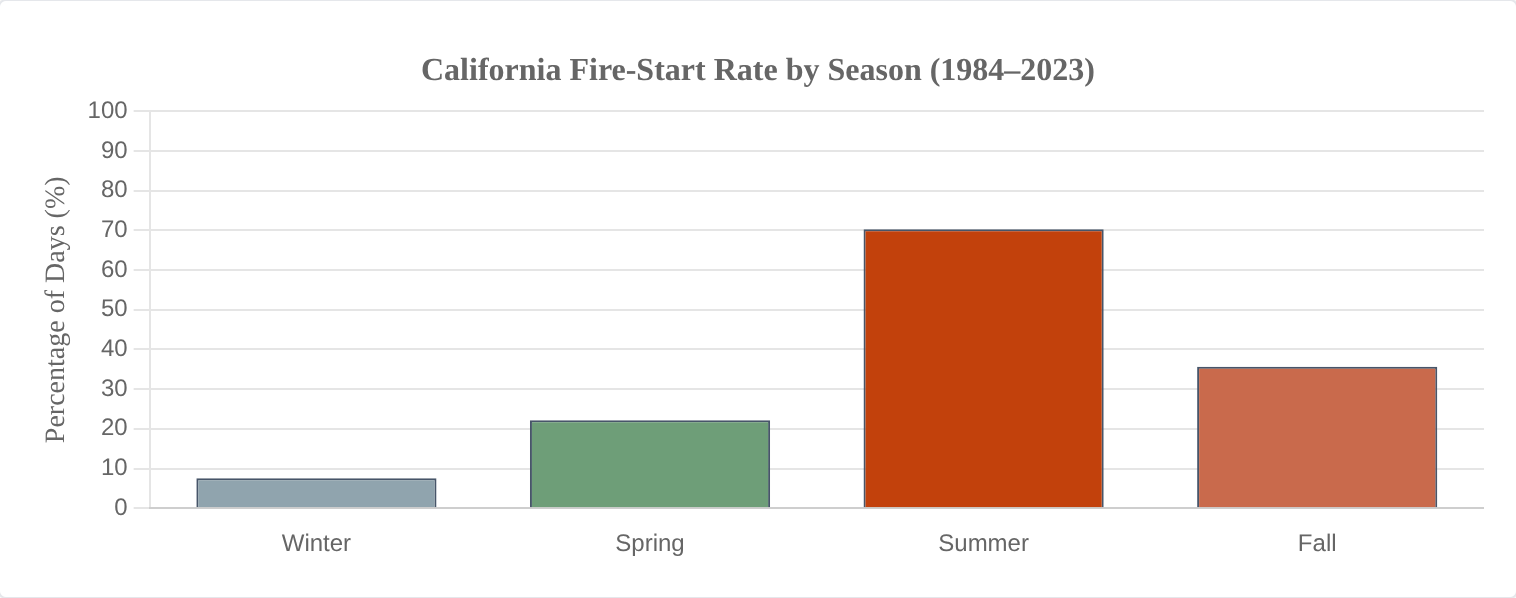
CRITICAL COVERAGE DECISION: 2024–2025 EXCLUSION

Audit revealed **zero positive labels** for the entire 2024 calendar year and early 2025, despite the positive labels present in prior years. To prevent contamination of the modeling-ready set with potential label-coverage gaps, these 378 records were excluded. Analysis focuses on the 1984–2023 period with consistent positive-label coverage.

Pedagogical Note: Detecting label gaps is a key data quality step.

Fire-start labels are seasonal

How does the timing of observations influence the target variable?



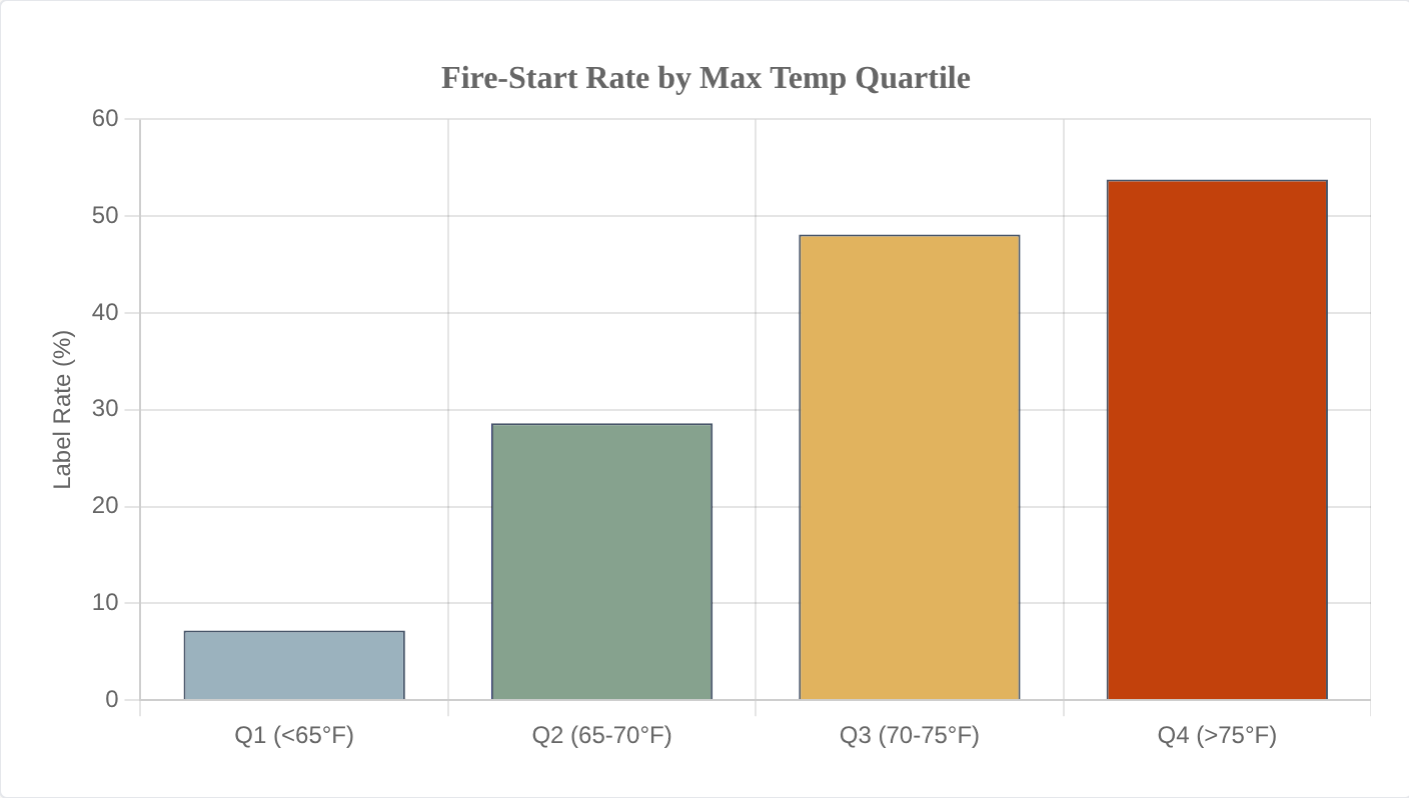
Analysis of the 1984–2023 period reveals a dominant seasonal cycle. **Summer** experiences fire-start labels on **70.3%** of days, whereas **Winter** sees them on only **7.5%** of days.

This pattern suggests seasonal features may be informative for a later model. However, seasonality is a *descriptive association*—season reflects complex meteorological and human activity patterns, not a direct cause.

Metric: Average Daily Fire-Start Rate (%) per Season. Data: 1984–2023.

Weather relationships in the sample

How does daily temperature volatility relate to wildfire occurrences?



Fire-start rates show a non-linear increase across temperature quartiles. In days with maximum temperatures above **75°F (Q4)**, the label rate reaches **53.8%**.

Conversely, in the lowest quartile (below 65°F), fire starts are recorded in only **7.2%** of observations.

Caveat: This does not establish 75°F as a physical trigger; it identifies temperature as a strong statistical associate in this specific historical record.

Quartiles: Q1 (<65°F), Q2 (65–70°F), Q3 (70–75°F), Q4 (>75°F). Data: 1984–2023.

Correlation: useful, not causal

Which numeric features align most strongly with fire-start labels?

- **Primary Predictors**

  Minimum and Maximum temperatures show the strongest positive linear correlation, aligning with seasonal heat intensity.

- **Negative Association**

  Lagged precipitation has a negative correlation (-0.21), so wetter preceding weeks coincide with fewer recorded fire-start labels in this sample.

- **Interpretative Limit**

  Correlation describes linear pairwise association only. It ignores interaction effects and non-linear patterns essential for ML.

Feature	Pearson Correlation
MIN_TEMP	+0.474
MAX_TEMP	+0.371
YEAR	+0.158
MONTH	+0.146
DAY_OF_YEAR	+0.144
PRECIPITATION	-0.116
LAGGED_PRECIPITATION	-0.214

Method: Pearson correlation with binary target. Period: 1984–2023.

Feature engineering choices

How is the raw environmental data transformed for the model?

Weather Baseline

Direct observations of Daily Max/Min Temp, Precipitation, and Wind Speed are retained as fundamental fire-weather proxies.

RAW INPUT

TEMP_RANGE & RATIO

Engineered features capturing daily thermal volatility and wind-to-heat intensity (WIND_TEMP_RATIO) to reveal non-obvious stress patterns.

ENGINEERED

LAGGED_PRECIP

A 7-day rolling precipitation window representing cumulative fuel moisture—an essential suppressor of ignition risk.

ENGINEERED (TIME)

SEASON & DATE

SEASON is one-hot encoded for categorical handling. Raw DATE is dropped to prevent overfit, but YEAR/MONTH/DAY_OF_YEAR are kept for trends.

ENCODED

Note: Raw labels (FIRE_START_DAY) and primary DATE are excluded from model inputs.

Preprocessing without leakage

How do we ensure future data doesn't "leak" into the training process?

1. **Sort & Split**
   Data sorted by DATE. First 80% assigned to Train, final 20% to Test.
2. **Learn Params**
   Fit Imputer & Scaler on **Train only**. Zero sight of Test values.
3. **Encode**
   Apply One-Hot Encoding to SEASON based on Train categories.
4. **Transform**
   Apply learned scaling/imputation to both Train and Test sets.

NO DATA LEAKAGE: Chronological split prevents "seeing the future." Scalers never see Test data statistics during the fitting stage.

Milestone 3 readiness checks

Is the dataset genuinely ready for Machine Learning models?

Final Split Metric	Train (1984-2015)	Test (2016-2023)
Rows	11,688	2,922
Features (Encoded)	15 Columns	15 Columns
Missing Cells	0	0
Positive Rate	31.03%	45.99%*

*Note: The held-out period has a higher positive-label rate; report this temporal distribution shift.

- ✓ Fully Numeric (X_train/X_test)
- ✓ Zero Preprocessing Leakage
- ✓ Strict Chronological Split
- ✓ Verified Imputation Strategy
- ✓ All Visualizations Documented

Deliverables: Clean Dataset, 2x Notebooks, Pipeline Artifacts, 4x Visuals.